SON 2383: Automated Parameter Estimation and Uncertainty Analysis for Water Quality Modeling

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Problem:

CE-QUAL-W2 is the USACE standard for reservoir water quality simulation, used by Districts nationwide to evaluate proposed projects, assess operational alternatives, and predict outcomes for temperature, nutrients, dissolved oxygen, and algal dynamics, including new capabilities for simulating and mitigating Harmful Algal Blooms (HABs) through, e.g., selective withdrawal, hypolimnetic aeration, and algal harvesting. However, model calibration remains manual, time-intensive, and subjective, producing inconsistent results and unquantified uncertainty that compromise the scientific defensibility of project evaluations.

Envisioned Product:

A modular Python toolkit that interfaces with CE-QUAL-W2 to automate calibration using optimization algorithms and quantify predictive uncertainty through Markov Chain Monte Carlo methods. The toolkit will include automated discovery of model configurations, parallel processing for high-performance computing environments, and diagnostic visualization modules. Deliverables include the open-source software, a technical report, a training suite of videos and webinars, and a pilot study on an active USACE project to validate enterprise readiness.

Value Added:

Automated calibration transforms labor-intensive trial-and-error tuning into a standardized process, reducing calibration timelines from months to days while introducing formal uncertainty quantification. It will provide planners with statistically derived confidence intervals rather than single-point predictions enables risk-informed decision-making for applications ranging from temperature modeling for endangered species to nutrient management for HAB mitigation.

****This accelerates pattern recognition and data integration allowing scientist to focus on interpretation, forecasting, and management decisions, which strengthens the workforce****

Regional/National Significance:

CE-QUAL-W2 has over 1,100 documented applications worldwide and is used by Districts across all USACE Divisions for reservoir water quality simulation. Every District that operates a CE-QUAL-W2 model faces the same calibration bottleneck; the toolkit eliminates it with a single standardized solution. Automated uncertainty quantification provides reproducible, defensible modeling results required for restoration and mitigation decisions, reducing legal and regulatory risk across the Civil Works portfolio. The framework is extensible to other USACE numerical models facing similar calibration challenges.